

# Stacking cell evaluation & gate audit

baseline (v1) at n=50 · commit 69d6830 · mujoco 3.11.0 · tip probability 0.2

SUCCESS RATE (N=50)

## 10%

5 of 50 episodes succeeded.  
The honest range around  
that number (95%  
confidence interval, Wilson):  
**4%–21%**

In other words: 50 episodes  
can't tell a 4% policy from a  
21% one.

Production gate ( $\geq 90\%$  at  
n=50): **does not pass**

HOW HIGH THE STACKS  
GOT

## 5·13·26·6

5 episodes ended with all  
three cylinders stacked, 13  
with two, 26 with one, 6 with  
none. Partial stacks are  
counted here, on their own —  
they are never mixed into the  
headline success rate.

COST

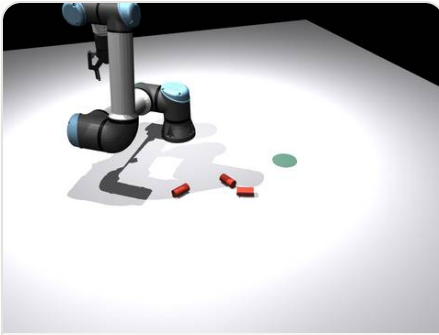
## 1s

seconds of real time per  
episode — the full 50-  
episode run took 48 seconds  
on a laptop

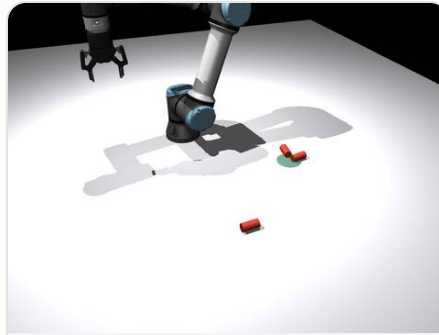
## Why episodes failed, ranked (45 failures)

|                                                                                           |                        |    |
|-------------------------------------------------------------------------------------------|------------------------|----|
| tipped part skipped                                                                       | <div><div></div></div> | 28 |
| a cylinder spawned lying down; the policy has no strategy for tipped parts and skipped it |                        |    |
| stack collapse                                                                            | <div><div></div></div> | 6  |
| placements destroyed the stack — one cylinder left standing at best                       |                        |    |
| grasp miss                                                                                | <div><div></div></div> | 5  |
| the gripper closed but captured nothing (approach missed the part)                        |                        |    |
| top placement failed                                                                      | <div><div></div></div> | 5  |
| two stacked cleanly; the third placement failed or knocked itself off                     |                        |    |
| base misplaced                                                                            | <div><div></div></div> | 1  |
| everything was lifted and placed, but no cylinder ended within tolerance of the target    |                        |    |

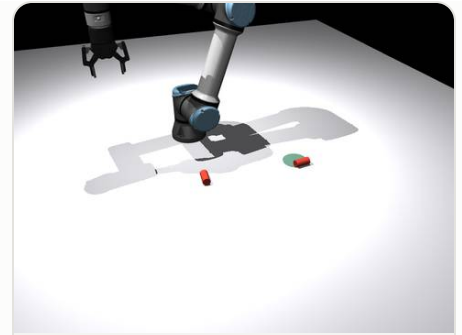
**The worst episodes, first — each one replayable with a single command**



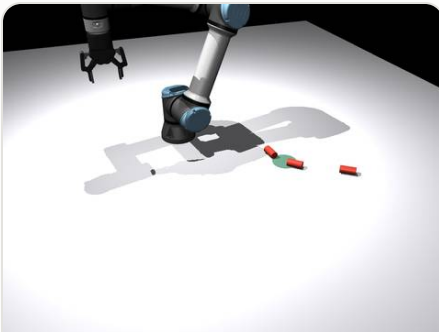
**seed 2** — stacked 0 of 3 —  
tipped part skipped  
watch it happen: `python  
render_video.py 2 --policy  
v1`



**seed 6** — stacked 0 of 3 — grasp  
miss  
watch it happen: `python  
render_video.py 6 --policy  
v1`



**seed 9** — stacked 0 of 3 —  
tipped part skipped  
watch it happen: `python  
render_video.py 9 --policy  
v1`



**seed 41** — stacked 0 of 3 — base  
misplaced  
watch it happen: `python  
render_video.py 41 --  
policy v1`

## The gate audit

Production rule under test: ship if  $\geq 45$  of 50 episodes succeed.

The policies in this audit:

- **v1** — the scripted baseline; it skips tipped-over parts
- **v2** — v1 plus a maneuver that stands tipped parts up first
- **mild / heavy** — v1 seeing the cylinders through 2 mm / 5 mm of seeded perception error
- **calibrated reference policies** — instruments with a dialed success rate, used in the near-threshold strips further down; labeled wherever they appear

## What 50 episodes said vs. what 1,000 revealed

| POLICY            | SMALL-SAMPLE MEASURED | TRUE RATE (N=1000) | 95% CI    |
|-------------------|-----------------------|--------------------|-----------|
| v1                | 10% (n=50)            | <b>5.3%</b>        | 4.1%–6.9% |
| v2                | 10% (n=50)            | <b>6.6%</b>        | 5.2%–8.3% |
| mild (2mm noise)  | 6% (n=100)            | <b>5.7%</b>        | 4.4%–7.3% |
| heavy (5mm noise) | 2% (n=100)            | <b>1.7%</b>        | 1.1%–2.7% |

- The gate-sized sample overestimated v1 **2×** (10% vs 5.3%).
- v2 trends better than v1 (6.6% vs 5.3%) but **even n=1000 does not resolve them** — their CIs overlap, as do the mild-noise variant's.
- Three mechanically different policies whose success counts never separate: the gate is mechanism-blind; the failure taxonomy above is what tells them apart.

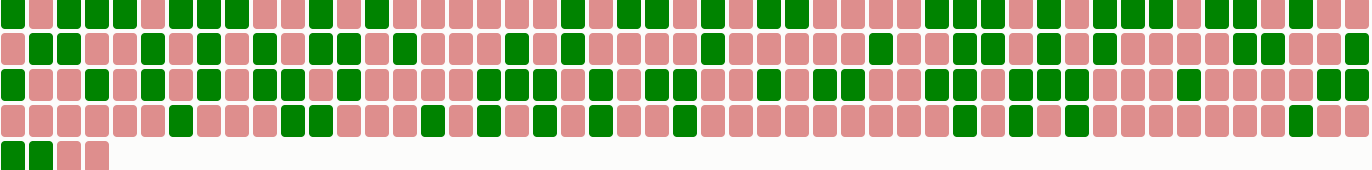
## Two policy variants, side by side — every gate verdict

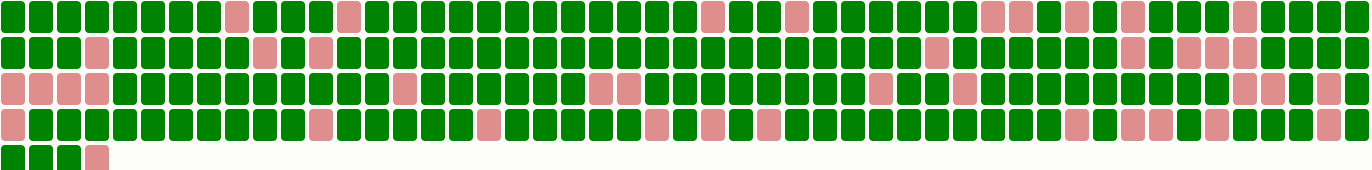
  
v1 — true rate 5.3% — 20 disjoint n=50 gates — 0/20 gates passed

  
v2 — true rate 6.6% — 20 disjoint n=50 gates — 0/20 gates passed

Far from the threshold the gate is reliable: two below-spec policies, 40 gates, zero false passes. The gate's errors live near the threshold — probed below with the calibrated reference policies introduced above.

## Watch the gate flip — calibrated policies near the threshold

  
calibrated reference policy, true rate 88% — measured pass rate 41.0% (exact theory: 43.5%) — 82/200 gates passed

  
calibrated reference policy, true rate 92% — measured pass rate 79.5% (exact theory: 79.2%) — 159/200 gates passed

**A below-spec 88% policy ships ~4 times in 10. A policy at exactly 90% spec is rejected ~4 times in 10.** Each square is a full 50-episode gate on its own seeds, judged episode by episode.

## Measured vs. theory, all levels

| TRUE RATE | PIPELINE (200 GATES) | COIN-FLIP (20K GATES) | EXACT THEORY | DISPERSION RATIO |
|-----------|----------------------|-----------------------|--------------|------------------|
| 80%       | 6.0%                 | 4.9%                  | 4.8%         | 1.04             |
| 85%       | 25.0%                | 21.4%                 | 21.9%        | 0.98             |
| 88%       | 41.0%                | 43.4%                 | 43.5%        | 1.08             |
| 90%       | 64.5%                | 62.2%                 | 61.6%        | 1.12             |
| 92%       | 79.5%                | 79.0%                 | 79.2%        | 0.94             |
| 95%       | 96.0%                | 96.3%                 | 96.2%        | 1.03             |

- Instrument calibration: dialed 88%, pipeline measured **88.4%** over n=3000 judged episodes.
- Scope, stated plainly: the reference policy draws each outcome from a seeded rate and arranges the scene for the real judge to rule — the sweep validates the **measurement chain** (judge, gate arithmetic, seeding) against a known truth, and cannot disagree with the coin-flip model unless the judge misrules an arranged scene.
- The gate's error rates therefore rest on the binomial model plus its independence assumption — checked on the **real** policies (dispersion ratios  $\approx 1$ : v1 0.93, v2 1.06, mild 0.99, heavy 1.17), not assumed.

## Reproduce

Every number regenerable from a fresh clone — commands in the README. Declared blind spot: grasping uses a kinematic attach within 20 mm; grasp-slip and drop-in-transit failures cannot occur in this evaluation.